

Asclepius Engineering Reports

The following engineering reports were generated by the Asclepius autonomous reasoning module following completion of the spacecraft health monitoring pipeline. For each mission scenario, detected fault events produced by the hybrid rule-based and machine learning anomaly detection framework were supplied to a locally hosted Llama 3.1 large language model together with mission-specific operational context. The model was instructed to produce structured flight operations reports containing an executive mission health assessment, prioritised fault ranking, distinction between confirmed failures and uncertain anomalies, subsystem interaction analysis, likely engineering causes, recommended operator actions, and an overall mission outlook. These reports demonstrate how large language models can transform low-level telemetry events into concise, evidence-based engineering assessments capable of supporting spacecraft operators during anomaly investigation and operational decision-making.

Earth Observation Mission:

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MISSION REASONING REPORT

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****Engineering Report: Earth Observation Mission****

****Executive Summary****

Mission Scenario: Earth Observation (Low Earth Orbit)

Overall Mission Health: ****Degraded****

Overall Mission Risk: ****Medium****

Number of Confirmed Failures: 5

Number of Uncertain Anomalies: 6

Spacecraft State: AOCS accuracy degradation, power and thermal issues, and communication signal degradation compromise the spacecraft's ability to perform imaging missions.

****Mission Health Assessment****

The spacecraft is experiencing significant degradation in multiple critical subsystems. The primary concerns are:

1. Attitude accuracy degradation (AOCS): High severity, high confidence
2. Power system degradation: Critical severity, high confidence
3. Communication signal degradation: Warning severity, high confidence

These issues compromise the spacecraft's ability to maintain its intended orbit and communicate with ground stations.

****Priority Assessment****

Ranking events from highest to lowest operational priority:

1. ****AOCS accuracy degradation**** (Subsystem: AOCS, Event: 'Attitude accuracy degradation', Duration: 136, Confidence: High, Severity: Critical)
* Prioritised due to critical severity and high confidence.
2. ****Power system degradation**** (Subsystem: power, Event: 'Voltage degradation', 'Current overload', 'Solar array degradation', Duration: 431, Confidence: High, Severity: Critical)
* Prioritised due to critical severity and high confidence.
3. ****Communication signal degradation**** (Subsystem: communication, Event: 'Signal degradation', 'Data rate reduction', 'Packet loss spike', Duration: 520, Confidence: High, Severity: Critical)
* Prioritised due to critical severity and high confidence.
4. ****Camera overheating**** (Subsystem: optics, Event: 'Camera overheating', Duration: 323, Confidence: High, Severity: Critical)
* Prioritised due to critical severity and high confidence.
5. ****Thruster overheating**** (Subsystem: propulsion, Event: 'Thruster overheating', 'Extreme thrust output', Duration: 141, Confidence: High, Severity: Critical)
* Prioritised due to critical severity and high confidence.

****Confirmed Failures vs Uncertain Anomalies****

Confirmed failures:

1. AOCS accuracy degradation
2. Power system degradation
3. Communication signal degradation
4. Camera overheating
5. Thruster overheating

Uncertain anomalies requiring monitoring:

1. Unknown anomalies in AOCS, optics, and thermal subsystems (Subsystems: AOCS, Optics, Thermal, Event: 'Unknown anomaly', Confidence: Medium)
2. OBC CPU overload (Subsystem: obc, Event: 'CPU overload', Duration: 76, Confidence: High, Severity: Warning)

These events are uncertain because they do not provide sufficient information to classify them as confirmed failures.

****Subsystem Interaction Assessment****

There is no conclusive evidence of interactions between subsystem events. However, power degradation could potentially contribute to communication signal degradation if the spacecraft's solar arrays cannot generate enough power for the communication system.

****Likely Causes****

1. AOCS accuracy degradation: likely caused by reaction wheel saturation or gyroscope drift anomaly (Subsystems: AOCS, Event: 'Reaction wheel saturation', 'Gyroscope drift anomaly')
 - * Hypothesis requiring investigation.
2. Power system degradation: possible causes include solar array degradation or power management unit failure (Subsystem: power, Event: 'Solar array degradation', 'Voltage degradation')
 - * Evidence-based conclusion and hypothesis requiring investigation.
3. Communication signal degradation: likely caused by signal degradation or data rate reduction (Subsystems: communication, Event: 'Signal degradation', 'Data rate reduction')
 - * Hypothesis requiring investigation.

****Recommended Operator Actions****

1. ****Reduce payload activity****: to conserve power and reduce thermal load on the spacecraft (Confidence: High)
2. ****Switch communication modes****: to optimise data transmission and reception (Confidence: Medium)
3. ****Enter safe power mode****: if power degradation is severe, to prevent further damage to the spacecraft's systems (Confidence: Low)

****Overall Mission Outlook****

Immediate operational risks:

1. AOCS accuracy degradation: may cause the spacecraft to drift out of its intended orbit.
2. Power system degradation: may lead to communication signal degradation and loss of data transmission.

Potential cascading failures:

1. Camera overheating: may damage the camera's electronics, leading to further payload availability issues.
2. Thruster overheating: may cause propulsion system failure, compromising the spacecraft's ability to maintain its orbit.

Most important subsystem to monitor next: AOCS, as its accuracy degradation has significant implications for the spacecraft's overall performance and stability.

Lunar Exploration Mission:

MISSION REASONING REPORT

Mission Report: Lunar Exploration Mission

Executive Summary

- * Mission scenario: Lunar Exploration
- * Overall mission health: Degraded
- * Overall mission risk: High
- * Number of confirmed failures: 8
- * Number of uncertain anomalies: 6
- * Brief summary: The spacecraft is experiencing a high level of subsystem degradation, with critical events affecting thermal survival, power availability, and communication availability.

Mission Health Assessment

The spacecraft's health is significantly degraded due to the cumulative impact of multiple subsystem failures. The thermal system has experienced prolonged overheating episodes, compromising temperature control. Power availability is also compromised, with solar array degradation leading to reduced energy production. Communication availability is severely impacted by signal degradation, antenna failure, and packet loss spikes.

Priority Assessment

The ranked events are as follows:

1. **Subsystem: AOCS, Event: Reaction wheel saturation (Critical)**

Duration: 467

Confidence: High

Event severity: Critical

Why it is prioritised: This event directly affects mobility operations and has high priority due to its critical severity.

2. **Subsystem: Communication, Event: Signal degradation & Antenna failure (Critical)**

Duration: 396

Confidence: High

Event severity: Critical

Why it is prioritised: This event severely impacts communication availability, which is a mission-critical function.

3. **Subsystem: Power, Event: Voltage degradation, Current overload, Solar array degradation (Critical)**

Duration: 458

Confidence: High

Event severity: Critical

Why it is prioritised: This event affects power availability and has high priority due to its critical severity.

4. **Subsystem: Thermal, Event: Bus overheating, Battery overheating, Payload overheating (Critical)**

Duration: 479

Confidence: High

Event severity: Critical

Why it is prioritised: This event affects thermal survival, which is essential for the mission's success.

5. **Subsystem: Communication, Event: Signal degradation & Data rate reduction & Packet loss spike (Critical)**

Duration: 469

Confidence: High

Event severity: Critical

Why it is prioritised: This event severely impacts communication availability and has high priority due to its critical severity.

6. **Subsystem: Propulsion, Event: Thruster overheating & Extreme thrust output (Critical)**

Duration: 224

Confidence: High

Event severity: Critical

Why it is prioritised: This event directly affects mobility operations and has high priority due to its critical severity.

****Confirmed Failures vs Uncertain Anomalies****

Confirmed failures:

- * Reaction wheel saturation (AOCS)

- * Signal degradation & Antenna failure (Communication)

- * Voltage degradation, Current overload, Solar array degradation (Power)

- * Bus overheating, Battery overheating, Payload overheating (Thermal)

- * Thruster overheating & Extreme thrust output (Propulsion)

Uncertain anomalies requiring monitoring:

- * Unknown anomaly in Communication subsystem

- * CPU overload, Memory saturation, and Memory integrity degradation in OBC subsystem
- * Solar array degradation, Voltage degradation, and Current overload in Power subsystem
- * Bus overheating and Payload overheating in Thermal subsystem

****Subsystem Interaction Assessment****

There is evidence of a plausible relationship between power degradation (event 4) contributing to communication degradation (events 1-3). Furthermore, the OBC overload (event 5) could potentially affect spacecraft autonomy.

****Likely Causes****

- * ****AOCS reaction wheel saturation****: Possible cause: thermal stress on reaction wheels.
- * ****Communication signal degradation & antenna failure****: Possible cause: solar array degradation leading to reduced energy production and compromised communication system functionality.
- * ****Power voltage degradation, current overload, solar array degradation****: Likely cause: solar array degradation causing reduced power generation and subsequent overloading of the power subsystem.
- * ****Thermal bus overheating, battery overheating, payload overheating****: Possible cause: electrical or thermal issues affecting temperature control.

****Recommended Operator Actions****

1. ****Enter survival mode****: Reason: The spacecraft's thermal system is critically degraded, and entering survival mode may help mitigate further damage.
 - * Confidence in recommendation: High
2. ****Prioritise communication windows****: Reason: Communication availability is severely impacted by signal degradation and antenna failure.
 - * Confidence in recommendation: High

****Overall Mission Outlook****

Immediate operational risks:

- * Continued thermal stress leading to increased risk of permanent damage.
- * Cascading failures affecting critical subsystems (power, communication).

Most important subsystem to monitor next: ****Power****.

Deep Space Mission:

MISSION REASONING REPORT

Executive Summary

Mission Profile: Deep Space Mission

Overall Mission Health: Critical

Overall Mission Risk: High

Number of Confirmed Failures: 8

Number of Uncertain Anomalies: 5

Brief Summary of the Spacecraft State:

The spacecraft is experiencing a high number of critical failures across multiple subsystems, including AOCS, power, and communication. These events are causing significant impact on mission objectives.

Mission Health Assessment

The overall condition of the spacecraft is Critical due to the numerous failures occurring across different subsystems. The AOCS system is particularly concerning with reaction wheel saturation and gyroscope drift anomalies impacting the spacecraft's ability to maintain its position and pointing accuracy. Power degradation is also a critical issue, as voltage degradation and solar array degradation are compromising the power generation capabilities of the spacecraft.

Priority Assessment

1. **AOCS: Gyroscope drift anomaly (1686-1691)**

- * Subsystem: AOCS
- * Event: Gyroscope drift anomaly
- * Duration: 5 minutes
- * Confidence: High
- * Severity: Warning
- * Why it is prioritised: Critical for spacecraft stability and control.

2. **Power: Voltage degradation, Solar array degradation (3301-3759)**

- * Subsystem: Power
- * Event: Voltage degradation, Solar array degradation

- * Duration: 458 minutes
- * Confidence: High
- * Severity: Critical
- * Why it is prioritised: Impacts power generation capabilities.

3. **AOCS: Reaction wheel saturation (1042-1509)**

- * Subsystem: AOCS
- * Event: Reaction wheel saturation
- * Duration: 467 minutes
- * Confidence: High
- * Severity: Critical
- * Why it is prioritised: Impacts spacecraft stability and control.

4. **Communication: Signal degradation, Antenna failure (1757-2153)**

- * Subsystem: Communication
- * Event: Signal degradation, Antenna failure
- * Duration: 396 minutes
- * Confidence: High
- * Severity: Critical
- * Why it is prioritised: Impacts communication capabilities.

5. **OBC: CPU overload (2563-2760)**

- * Subsystem: OBC
- * Event: CPU overload
- * Duration: 197 minutes
- * Confidence: High
- * Severity: Critical
- * Why it is prioritised: Impacts computer reliability and data processing capabilities.

6. **Communication: Signal degradation, Data rate reduction, Packet loss spike (3541-4010)**

- * Subsystem: Communication
- * Event: Signal degradation, Data rate reduction, Packet loss spike
- * Duration: 469 minutes
- * Confidence: High
- * Severity: Critical
- * Why it is prioritised: Impacts communication capabilities.

7. **Power: Solar array degradation (164-173)**

- * Subsystem: Power
- * Event: Solar array degradation
- * Duration: 9 minutes
- * Confidence: High
- * Severity: Warning
- * Why it is prioritised: Early sign of power generation issues.

8. ****Communication: Unknown anomaly (3282-3294)****

- * Subsystem: Communication
- * Event: Unknown anomaly
- * Duration: 12 minutes
- * Confidence: Medium
- * Severity: Investigation
- * Why it is prioritised: Requires further investigation due to unknown nature.

****Confirmed Failures vs Uncertain Anomalies****

1. Confirmed failures:

- * AOCS: Reaction wheel saturation (1042-1509)
- * Power: Voltage degradation, Solar array degradation (3301-3759)
- * OBC: CPU overload (2563-2760)
- * Communication: Signal degradation, Antenna failure (1757-2153)

These events have been confirmed by multiple detectors and/or ML analysis.

2. Uncertain anomalies:

- * Communication: Unknown anomaly (3282-3294)
- * Power: Unknown anomaly (1737-1743)
- * OBC: CPU overload (2775-2909)

These events require further investigation due to insufficient evidence or conflicting detector results.

****Subsystem Interaction Assessment****

Possible relationships between subsystem events are as follows:

- * Power degradation contributing to communication degradation. (Confirmed by multiple signal degradation events)
- * AOCS degradation affecting communication pointing accuracy. (Plausible due to shared dependencies on power and thermal systems)

These interactions require further investigation to establish causality.

****Likely Causes****

Based on event data, the following likely causes are proposed:

1. ****AOCS: Gyroscope drift anomaly**** - Possible reason: Software issue or hardware malfunction affecting gyroscope performance.
2. ****Power: Voltage degradation**** - Possible reason: Solar array degradation or other power generation issues affecting voltage levels.

These hypotheses require further investigation to establish causality.

****Recommended Operator Actions****

1. ****Activate redundant systems (AOCS, Power)****
 - * Reason: Compensate for degraded subsystems and ensure continued mission operations.
 - * Confidence: High
2. ****Perform onboard diagnostics (OBC, Communication)****
 - * Reason: Identify root causes of CPU overload and communication signal degradation.
 - * Confidence: Medium
3. ****Delay non-critical operations****
 - * Reason: Prioritize essential tasks while investigating critical subsystem failures.
 - * Confidence: High

These recommendations prioritize the most critical subsystems and events.

****Overall Mission Outlook****

Immediate operational risks:

- * AOCS stability and control
- * Communication capabilities
- * Power generation and management

Potential cascading failures:

- * Continued AOCS degradation impacting communication pointing accuracy
- * OBC overload affecting spacecraft autonomy

Most important subsystem to monitor next:

- * AOCS, due to critical implications on spacecraft stability and control.